AI-ENHANCED FAULT DETECTION AND SELF-HEALING IN OPTICAL FIBER NETWORKS

A M I N I PROJECT REPORT

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An Autonomous Institution | Approved by AICTE | Affiliated to Anna University | Accredited by NAAC with A++ Grade

Kuniamuthur, Coimbatore – 641008.

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SUSTAINABLE DEVELOPMENT GOALS

The Sustainable Development Goals are a collection of 17 global goals designed to blue print to achieve a better and more sustainable future for all. The SDGs, set in 2015 by the United Nations General Assembly and intended to be achieved by the year 2030, In 2015, 195 nations agreed as a blue print that they can change the world for the better. The project is based on one of the 17 goals.

Question	Answer
Which SDGs does the project directly address?	The project directly supports SDG 9: Industry, Innovation and Infrastructure by improving the reliability and intelligence of fiber communication systems.
What strategies or actions are being implemented to achieve these goals?	The project applies AI-based fault classification and intelligent rerouting algorithms to ensure continuous fiber-network connectivity.
How is progress measured and reported in relation to the SDGs?	Progress is measured by tracking reductions in network downtime, faster fault-detection time, and improved rerouting success rates.
How were these goals identified as relevant to the project's objectives?	These goals were chosen because the project focuses on improving communication reliability and strengthening digital infrastructure.
Are there any partnerships or collaborations in place to enhance this impact?	Yes, the project can be enhanced through collaboration with telecom service providers, fiber network operators, or academic research groups.

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BONAFIDE CERTIFICATE

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ABSTRACT

Optical fiber communication networks require continuous fault monitoring to ensure reliable and uninterrupted data transmission. Physical impairments such as fiber cuts, bending losses, and poor splices can severely distort the backscattered optical signal, affecting overall network performance. Optical Time Domain Reflectometry (OTDR) serves as a powerful technique for detecting such degradations by analyzing the variation of reflected signal power along the fiber. This project presents a MATLAB-based intelligent OTDR monitoring and self-healing system that simulates realistic OTDR traces, automatically identifies the presence and type of faults, and restores connectivity through dynamic rerouting. The system generates synthetic OTDR profiles for four conditions—Normal, Cut, Bend, and Splice—using a custom trace-generation model that accurately reflects attenuation patterns and fault-induced power deviations. Key signal features such as drop magnitude, slope change, and local variation are extracted and classified using a Decision Tree algorithm, chosen for its interpretability, fast execution, and suitability for threshold-based OTDR pattern recognition. Upon detecting a fault, the system initiates a network-self-healing module that applies Dijkstra's Shortest Path Algorithm to compute an optimal alternate communication route, thereby maintaining service continuity even under failure conditions. The integrated MATLAB GUI provides an interactive platform for visualization, analysis, fault interpretation, and dynamic rerouting, demonstrating an effective intelligent monitoring solution for modern optical fiber networks. Additionally, the system architecture ensures modular integration with existing network management frameworks, allowing seamless scalability. Real-time processing capabilities further enable immediate response to fault events, reducing network downtime. The synthetic OTDR engine is flexible and can be extended to include additional fault categories for advanced analysis.

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CHAPTER 1
INTRODUCTION

1.1 OVERVIEW

Optical fiber communication is a high-speed, high-bandwidth data transmission technology that uses light signals to carry information over long distances. Due to its low attenuation, immunity to electromagnetic interference, and high data-carrying capacity, optical fiber has become the backbone of modern telecommunication systems, including internet, broadband, military communication, and data centers. The quality of transmission in an optical fiber depends on the physical condition of the fiber cable, splices, connectors, and environmental factors.

1.2 NEED FOR FAULT DETECTION IN FIBER NETWORKS

As communication networks expand, fiber cables are deployed across long geographical areas and are exposed to environmental damage, construction work, bending stress, and aging. Even a small physical impairment can lead to significant signal loss or complete communication failure.

Therefore, continuous monitoring and early fault detection are essential to:

- Minimize network downtime
- Ensure service reliability
- Reduce maintenance cost
- Enable quick fault localization for field technicians
- Early detection improves system stability and helps maintain uninterrupted connectivity for critical applications.

1.3 ROLE OF OTDR IN FIBER MONITORING

Optical Time Domain Reflectometry (OTDR) is the most widely used diagnostic tool for fault detection in optical fibers. By sending optical pulses and measuring the backscattered signal, OTDR generates a trace representing the physical condition of the fiber.

OTDR helps to:

- Locate fiber breaks
- Detect bends and splice losses
- Measure return loss and attenuation
- Provide a visual representation of fiber health
- It is non-invasive, highly sensitive, and capable of identifying faults over long distances, making it essential for fiber network maintenance.

1.4 MOTIVATION FOR INTELLIGENT (AI-BASED) OTDR ANALYSIS

Traditional OTDR analysis depends on skilled technicians manually interpreting the trace patterns to identify fiber impairments. This approach is slow, prone to human error, and becomes increasingly inefficient as modern optical networks grow in size and complexity. Manual interpretation struggles with similar-looking fault signatures, inconsistent analysis between users, and delayed response times in large-scale networks. Therefore, there is a strong need for an automated and intelligent solution that can handle continuous monitoring without human intervention. AI-based OTDR analysis provides this advantage by automatically extracting features from OTDR traces, classifying faults such as Normal, Cut, Bend, and Splice, and determining their exact location. In addition, the system incorporates automated network rerouting using optimization algorithms, ensuring uninterrupted communication even when a fault occurs. This intelligent, fast, and consistent analysis significantly enhances network reliability, reduces downtime, and supports the increasing demands of high-speed optical communication systems.

1.5 PROBLEM STATEMENT

Optical fiber networks currently lack an integrated intelligent system capable of detecting, classifying, and responding to faults in real time. Manual OTDR interpretation is slow and error-prone, making it unsuitable for modern high-capacity networks that require rapid fault detection and recovery. Existing systems also do not offer a built-in mechanism to restore the communication path automatically when a fault is detected. As a result, delays in fault identification and network recovery lead to reduced performance and service interruptions. Therefore, the problem addressed in this project is the absence of:

- A real-time simulated OTDR monitoring environment
- Automatic fault detection and classification for Normal, Cut, Bend, and Splice conditions
- An intelligent decision-making model for analyzing OTDR features
- An autonomous mechanism for restoring connectivity through automated rerouting using shortest-path optimization algorithms
- This project aims to address these challenges through an intelligent MATLAB-based solution.

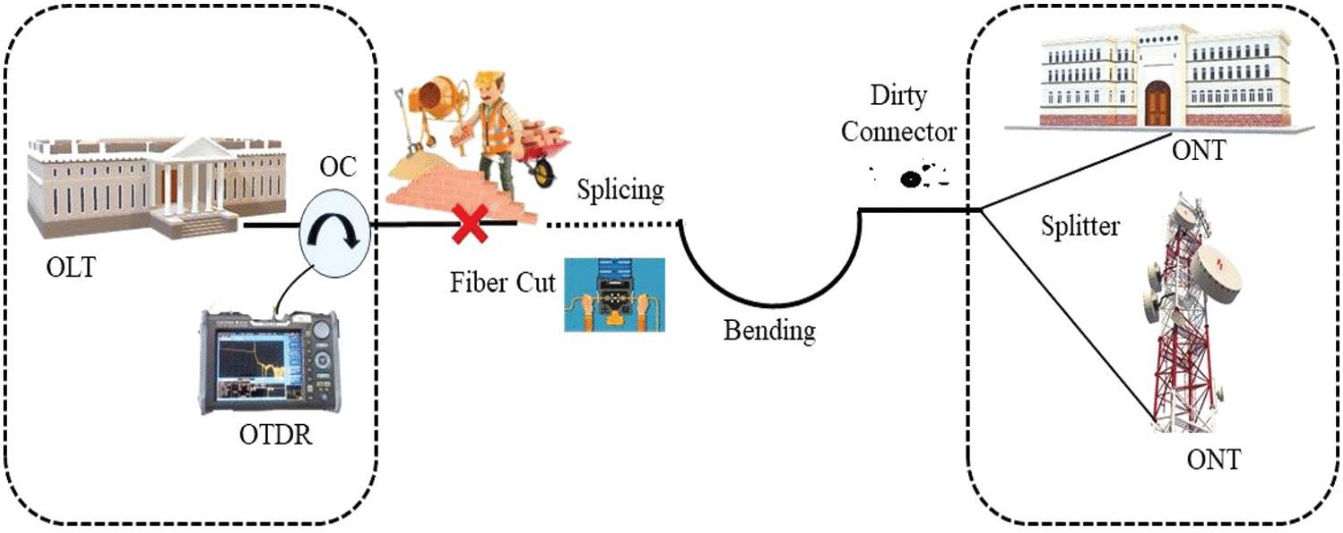


Fig 1.1 Problem Statement

1.6 SCOPE OF THE PROJECT

This project focuses on designing an intelligent OTDR-based fault detection and self-healing system using MATLAB. The scope includes the simulation of OTDR traces for various fault scenarios, extraction of relevant signal features, and classification of fault type using a Decision Tree algorithm. The system provides clear visualization of OTDR traces and fault locations through a MATLAB GUI. A key component of the project is the implementation of automated network rerouting using optimization algorithms, such as Dijkstra's Shortest Path Algorithm, to ensure uninterrupted communication after a fault is detected. The project is entirely software-based and does not involve any hardware implementation, but it establishes a strong foundation that can be extended to interface with real OTDR sensors and deployed in actual optical networks.

CHAPTER 2

PROBLEM DEFINITION & OBJECTIVES

2.1 PROBLEM DEFINITION

Optical fiber networks form the backbone of modern communication systems, carrying high-speed data over long distances. Despite their reliability, fiber links are susceptible to physical disturbances such as bending, cable cuts, and poor splicing. Identifying these issues typically requires OTDR trace analysis, which is complex and depends on expert interpretation. As networks grow in size and traffic load, traditional manual fault analysis becomes slow and inconsistent.There is a need for a system that can intelligently analyze OTDR traces, identify different fault types, determine their locations, and initiate a rerouting mechanism to maintain network connectivity when faults occur. This project aims to provide such an intelligent framework using MATLAB.

2.2 EXISTING SYSTEM LIMITATIONS

Existing OTDR-based monitoring systems have several limitations that reduce their effectiveness in maintaining reliable fiber communication networks. The major drawbacks are as follows:

1. Heavy Dependence on Human Expertise

Traditional OTDR systems require skilled technicians to observe and interpret the trace. Since the interpretation depends on individual experience, the results often vary, leading to inconsistency in identifying the exact fault type or its severity. In large networks, this dependency becomes increasingly difficult to manage due to the volume of data.

2. Slow and Time-Consuming Fault Identification

Manual inspection of OTDR traces takes considerable time, especially when long fiber spans or multiple fault points are involved. This delay increases the duration of network outages and affects the quality of service. As modern networks demand near-continuous uptime, slow fault detection becomes a major limitation

3. Difficulty in Differentiating Similar Fault Patterns

Some faults—such as bends, splice losses, and gradual attenuation—produce OTDR signatures that appear very similar. Existing systems often fail to distinguish them accurately. As a result, technicians may misclassify faults, leading to incorrect repair activities and repeated service disruptions.

4. Limited Analytical and Decision-Making Capabilities

Conventional OTDR machines primarily display the trace graph but do not provide intelligent analytical support. They lack integrated fault classification, pattern recognition, or explanation features. More importantly, they do not provide guidance or decision support for subsequent corrective actions.

5. No Built-In Mechanism to Maintain Network Flow During Faults

Most existing systems can detect that a fault has occurred but do not include any mechanism to ensure the continuous flow of communication when the fault is present. In large networks, the absence of a rerouting mechanism means that data transmission stops completely until the fault is manually repaired, reducing overall network reliability.

6. Not Suitable for Large-Scale Modern Fiber Networks

With increasing network size, higher bandwidth demand, and multiple nodes, traditional OTDR systems struggle to keep up. They lack scalability and cannot process or analyze data efficiently for complex topologies, making them unsuitable for modern intelligent communication infrastructures.

2.3 PROPOSED SYSTEM OVERVIEW

The proposed system is a MATLAB-based intelligent OTDR monitoring and analysis platform designed to overcome the limitations of traditional OTDR interpretation. Instead of relying on manual trace reading, the system generates simulated OTDR signals for multiple fiber conditions—including Normal, Cut, Bend, and Splice—and processes them using analytical and decision-making algorithms. These simulated traces form the basis for extracting meaningful features such as power drop magnitude, backscatter variation, slope changes, and attenuation characteristics. Using these extracted features, the system performs fault classification with the help of a Decision Tree model, which is well-suited for establishing clear, rule-based distinctions between different fault types. Once a fault is identified, the system computes an alternate communication path using Dijkstra's shortest path algorithm. This ensures that data transmission can continue even when a section of the fiber is damaged. A MATLAB App Designer GUI is included to make the entire workflow visual and user-friendly, allowing users to load traces, view classification results, observe fault locations, and see the rerouted network path—all within a single integrated environment. Although hardware is not implemented, the system follows real OTDR behavior and provides a realistic framework that can later be interfaced with physical sensors or real fiber-network data.

2.4 PROJECT OBJECTIVES

The primary objectives of this project are designed to build a complete and intelligent OTDR-based fiber monitoring system capable of detecting, classifying, and responding to faults in a simulated network environment. The key objectives include:

1. To generate realistic OTDR traces representing various fiber conditions such as Normal, Cut, Bend, and Splice using MATLAB simulation techniques.

2. To extract meaningful signal features (such as slope variation, minimum/maximum values, sudden power drops, attenuation changes, and deviation patterns) that help in distinguishing different fault types.

3. To classify fiber faults accurately using a Decision Tree algorithm, chosen for its interpretability and ability to create threshold-based rules that suit OTDR signal behavior.

4. To identify and highlight fault locations on the OTDR trace and visually represent the detected anomaly within the GUI.

5. To compute an alternate communication path using Dijkstra's Shortest Path Algorithm to maintain network connectivity when a fault is present in the fiber.

6. To design a complete and interactive MATLAB App Designer GUI that integrates all modules—trace generation, feature extraction, classification, visualization, and rerouting—within a single platform.

7. To provide clear explanations of the detected fault type and the reason behind its classification to support better understanding and transparency of the analysis.

8. To create a modular system architecture that can later be integrated with real OTDR hardware or external data sources without modifying the core algorithms.

9. To support educational and research applications by demonstrating how OTDR signal processing, machine learning, and routing algorithms work together in a unified fiber monitoring system.

With these objectives, the project aims to deliver a realistic, scalable, and easy-to-understand framework suitable for both simulation and potential future deployment.

2.5 EXPECTED OUTCOMES

The project is expected to deliver a MATLAB-based intelligent OTDR monitoring system capable of simulating Normal, Cut, Bend, and Splice faults with realistic trace behaviour. The system will extract key features from the OTDR signal and accurately classify the fault type using a Decision Tree model. It will identify and highlight the exact fault location on the plotted trace. The rerouting module based on Dijkstra's Algorithm will provide an alternative communication path whenever a fault is detected.

The final outcome will be a complete, user-friendly MATLAB App Designer interface that integrates trace generation, fault detection, classification, visualization, and network rerouting into a single unified tool.

CHAPTER 3

LITERATURE REVIEW

Optical fiber communication systems have become the backbone of modern telecommunication due to their high bandwidth, low loss, and immunity to electromagnetic interference. As networks continue to expand, maintaining service continuity requires rapid and accurate detection of physical defects within the fiber. This need has motivated researchers to develop enhanced monitoring methods, especially using Optical Time Domain Reflectometry (OTDR).Early research focused primarily on traditional OTDR analysis, where technicians interpret backscattered traces manually. Studies show that manual interpretation is often time-consuming and prone to inconsistencies, especially when detecting subtle defects such as micro-bends, weak splices, and gradual losses. These limitations encouraged the development of automated signal processing and feature extraction techniques to assist technicians.Subsequent works introduced mathematical and statistical approaches for analyzing OTDR traces. Researchers proposed gradient-based loss detection, curve fitting, thresholding methods, and segmentation techniques to improve detection accuracy.

CHAPTER 4

SYSTEM OVERVIEW

4.1 PROPOSED SYSTEM ARCHITECTURE

The proposed system combines OTDR-based fault detection, feature extraction, classification, and rerouting into one integrated MATLAB-based platform.

The architecture consists of three core layers:Signal Simulation Layer, which generates OTDR traces for Normal, Cut, Bend, and Splice conditions.Intelligent Analysis Layer, where fault features are extracted and processed by a Decision Tree classification model to determine the type and severity of the fault.Network Rerouting Layer, which uses Dijkstra's Shortest Path Algorithm to compute an alternate route when a fault is identified.These three layers are coordinated through a MATLAB App Designer interface that serves as the user interaction point. The architecture ensures smooth data flow—from trace generation, to analysis, to visualization, and finally to rerouting—making the system efficient and easy to use.

4.2 FRONTEND (MATLAB APP DESIGNER) OVERVIEW

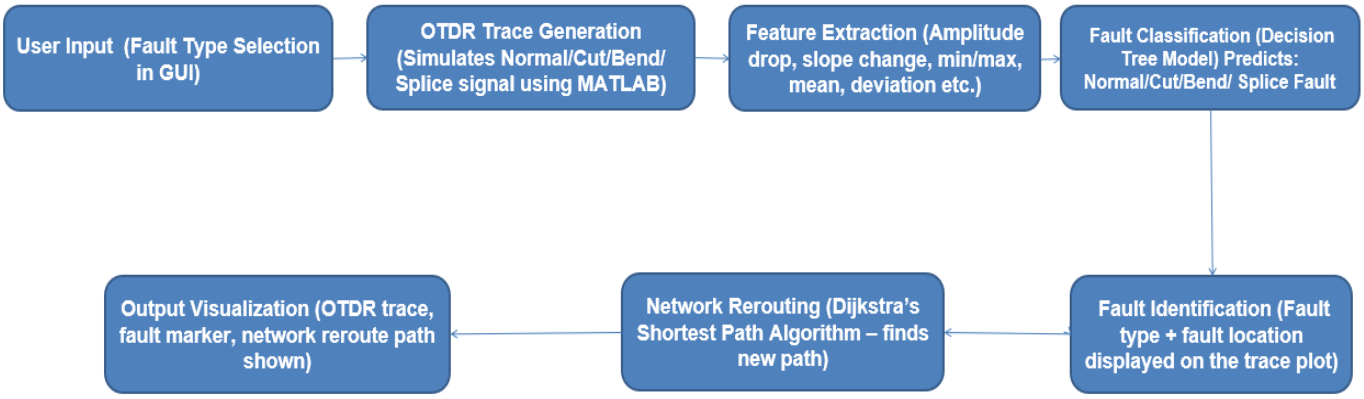
The system's frontend is developed using MATLAB App Designer, providing a clean and interactive graphical interface. Users can select the type of fault, view the generated OTDR trace, extract features, classify faults, and visualize the rerouted network. Major components of the GUI include:Dropdown Menu to select the fault type.Analyze Button to generate and display the OTDR trace.Feature & Explanation Button to compute features and display the classification result.Reroute Button to display the alternate network path.Axes Components to visualize fiber traces and network diagrams.The frontend ensures that even users without deep technical knowledge can easily interact with the system and understand the results.

4.3 BACKEND MODULES OVERVIEW

The backend is the computational core of the system and consists of the following modules:a) Trace Generation ModuleImplements the generate_traces() function that simulates realistic OTDR signals based on the selected fault type. It introduces attenuation, noise, and fault-based deviations.b) Feature Extraction ModuleExtracts mathematical features such as signal drop magnitude, slope changes, mean, minimum, and maximum values. These features help the classifier differentiate between Cut, Bend, Splice, and Normal conditions.c) Fault Classification ModuleUses a Decision Tree algorithm because of its simplicity, interpretability, and threshold-based decision rules that match OTDR characteristics.

The classifier predicts the fault type and provides an explanation.d) Rerouting ModuleImplements Dijkstra's Shortest Path algorithm to find an alternate route when a fault occurs.This module improves network continuity by bypassing the faulty segment and preserving communication flow.These backend modules work in synchronization to perform complete intelligent OTDR analysis.

4.4 BLOCK DIAGRAM



CHAPTER 5

CLASSIFICATION OF FIBER FAULTS

Optical fiber links can experience several types of physical faults, and each one creates a unique pattern in the OTDR trace. By analyzing the slope, sudden drops, and shape of the signal, it becomes possible to classify the condition of the fiber accurately. This chapter explains each major fault type with the corresponding OTDR behavior and the exact placement of images.

5.1 Normal Condition

A normal fiber shows a smooth and gradual decrease in signal strength caused by natural attenuation. The trace is linear without any abrupt variations.

Characteristics

Straight downward slope

No sudden jumps or drops

Slight random noise present

Indicates a fully healthy fiber

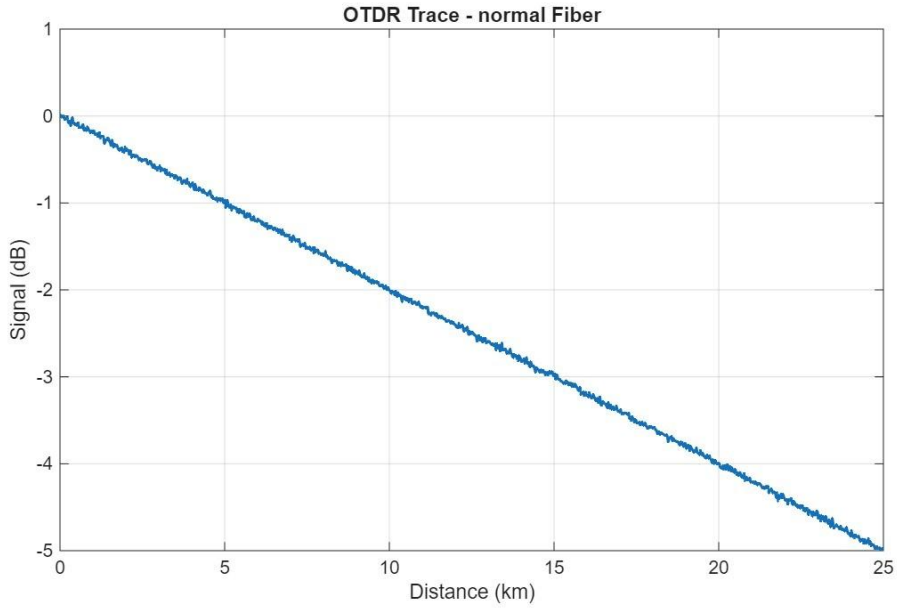


Fig 5.1 Normal Condition

5.2 CUT FAULT – SUDDEN SIGNAL DROP

A cut fault occurs when the fiber is physically broken. The OTDR trace shows a very sharp drop at the cut point, after which the signal becomes flat because no further backscatter is returned.

Characteristics

Sudden vertical drop at the cut location

Signal becomes flat after the cut

Indicates complete loss of connection beyond the break\

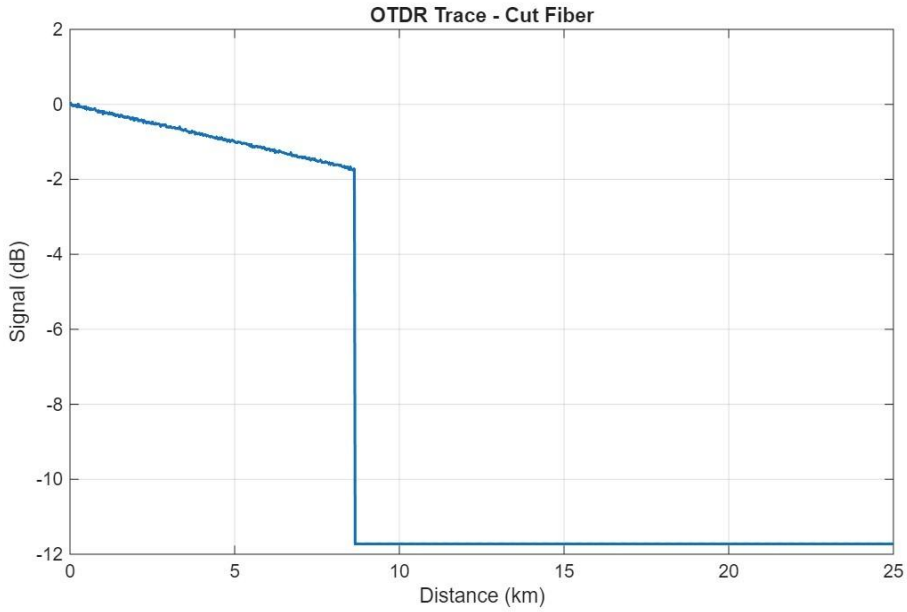


Fig 5.2 Cut - Fault

5.3 BEND FAULT – GRADUAL LOSS

A bend fault happens when the fiber is bent beyond its allowed radius. This causes increased attenuation, which appears as an extra slope or curved drop in the OTDR trace.

Characteristics

Slope becomes steeper after the bend point

Gradual drop instead of sharp fall

Indicates excessive bending or tight routing

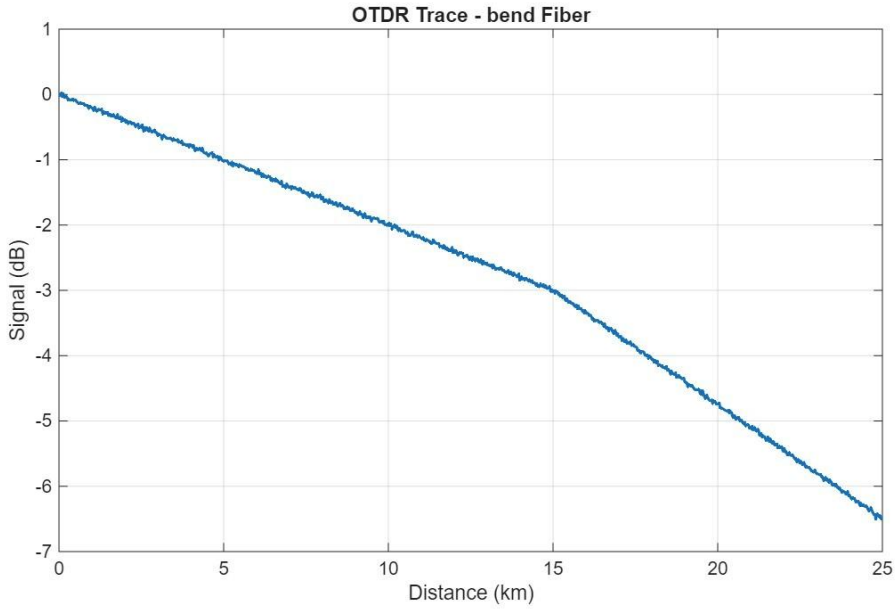


Fig 5.3 Bend - Fault

5.4 SPLICE FAULT – MINOR REFLECTION DROP

A splice fault occurs due to improper joining of two fiber ends. This causes a small step drop in the signal because of splice loss or bad alignment.

Characteristics

Small but noticeable drop in signal at splice point

Slope continues normally after the drop

Indicates poor splice quality or dust at joint

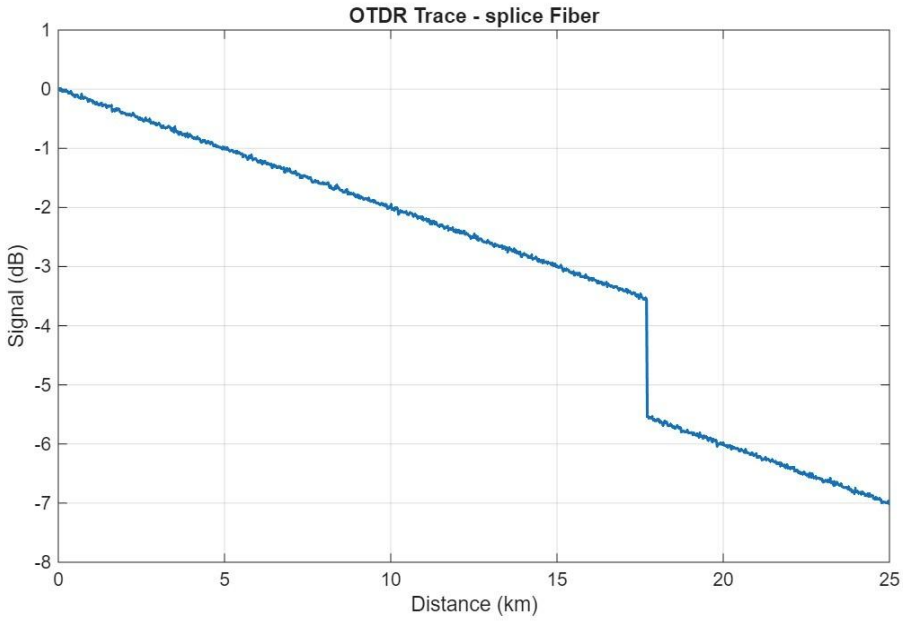


Fig 5.4 Splice - Fault

5.5 IMPORTANCE OF CLASSIFYING EACH FAULT TYPE

Accurate classification of fiber faults is essential for efficient maintenance and restoration of communication networks. Different faults require different corrective actions:Cut faults → require immediate physical repairBend faults → can be fixed by adjusting fiber routingSplice faults → need re-splicing or cleaningNormal condition → confirms that the network is healthyBy understanding the specific type of fault, technicians can reduce repair time, avoid unnecessary replacements, and improve network uptime.

5.6 INDIVIDUAL SIGNAL BEHAVIOR MODEL FOR EACH FAULT

Each fault type produces a unique signal pattern:

Normal fiber: Linear, low-loss attenuation

Cut fiber: Sudden drop followed by zero backscatter

Bend fiber: Smooth curve or increased slope after bend point

Splice fault: Small step loss at the splice locationThese distinct patterns allow advanced algorithms (signal analysis or machine learning) to classify faults automatically, ensuring faster diagnosis and improved monitoring of fiber networks.

CHAPTER 6
METHODOLOGY

6.1 SIGNAL SIMULATION USING GENERATE_TRACES.M

The generate_traces.m function is responsible for generating simulated OTDR signals for different fiber conditions such as Normal, Cut, Bend, and Splice. This function produces the distance-power trace, adds noise to replicate real OTDR measurements, and introduces fault-specific modifications to the signal. It also returns the location of the fault to assist with plotting and classification.

Explanation of the Function

1. InputfaultType – The type of fault to simulateOptions: "Normal", "Cut", "Bend", "Splice“

2. Output

- Distance – 0 to 10 km fiber length (1000 samples)
- Signal – OTDR backscatter power (db)
- Faultlocation – index where the fault occurs
- Isfaultdetected – boolean flag showing if a fault exists

3. Signal Generation Process

- A distance vector from 0–10 km is created.
- A baseline attenuation of −0.2 dB/km is applied.
- Gaussian noise is added to simulate real OTDR traces.

Based on the selected fault:

- Normal: No changes, no fault.
- Cut: Sudden drop of approx. −10 dB at a random location
- Bend: Gradual drop over ~20 samples (~small bend-induced loss)
- Splice: Small loss (~1.5 dB) over a smaller region.

Pseudo-Code for generate_traces.m

```
BEGIN
FUNCTION
generate_traces(faultType)
SET distance vector from 0 to 10 km with 1000 points
CREATE baseSignal = -0.2 * distance    // linear attenuation     ADD random noise to
baseSignal → signal
 INITIALIZE faultLocation = empty
INITIALIZE isFaultDetected = false
 SWITCH faultType
CASE "normal":
// Do nothing (no fault)
 isFaultDetected = false
CASE "cut":
 RANDOMLY choose a location between 200 and 900 → faultLocation
MAKE signal drop by 10 dB from this point forward
SET isFaultDetected = true
CASE "bend":
RANDOMLY choose a location → faultLocation
REDUCE signal by 3 dB for ~20 samples ahead
SET isFaultDetected = true
CASE "splice":
RANDOMLY choose a location → faultLocation
REDUCE signal by 1.5 dB for ~10 samples ahead
SET isFaultDetected = true
DEFAULT:
DISPLAY error: Unknown fault type
END SWITCH
RETURN distance, signal, faultLocation, isFaultDetected
END FUNCTION
END
```

6.2 FEATURE EXTRACTION USING EXTRACT_FEATURES.M

Feature extraction is a key step in analyzing OTDR signals because raw trace data cannot directly indicate fault type. The extract_features.m function converts the OTDR signal into meaningful numerical descriptors that help the classifier recognize patterns associated with Normal, Cut, Bend, or Splice conditions. The extracted features include statistical, gradient-based, and shape-based metrics, along with fault location estimation, making the classification more accurate and reliable.

6.2.1 MEAN

The mean of the signal gives the average power level across the entire fiber length.It helps detect global signal drops or weakening trends, which may occur due to large losses such as cuts or severe bends.

6.2.2 STANDARD DEVIATION

The standard deviation measures how widely the signal values vary from the mean.Faults like bends or splices distort the signal shape and introduce fluctuations, causing the standard deviation to increase compared to a normal trace.

6.2.3 MINIMUM VALUE

The minimum value represents the lowest backscattered power in the trace.This is often the fault indicator point, especially in a Cut fault where the signal sharply collapses.

6.2.4 MAXIMUM VALUE

The maximum value shows the highest recorded signal in the trace.This helps in understanding the trace's dynamic range and may indicate reflection peaks.

6.2.5 FAULT LOCATION

The fault location is estimated by identifying the index of the minimum signal value and mapping it to the corresponding distance in kilometers.Since most faults produce a sharp drop, this index serves as a simple and effective method to approximate where the fault occurs along the fiber.

Pseudo-Code for Feature Extraction

```
BEGIN
INPUT: distance array, signal array
CONVERT distance and signal to column vectors
// Statistical Features
Compute Mean of signal
Compute Standard Deviation
Find Minimum signal value
Find Maximum signal value
Compute Range = Max – Min
Compute Variance of signal
// Gradient-Based Features
Compute gradient = difference of signal / difference of distance
Compute AvgSlope = mean of gradient
Compute MaxSlope = maximum absolute gradient
// Shape-Based Features
Compute Skewness of signal
Compute Kurtosis of signal
// Fault Location Feature
Find index of minimum signal value
Map that index to distance value → FaultLocationKM
STORE all computed values in a feature structure
RETURN feature structure
END
```

6.3 FAULT PREDICTION USING PREDICT.M

The predict_fault function is responsible for simulating a new OTDR trace, extracting its numerical features, and classifying the fault type using the trained Decision Tree model.

The function first loads the saved model (trainedModel.mat), then generates a signal for the selected fault type, computes its features, and aligns them with the model's required predictor format. Finally, it performs the classification and plots the OTDR trace with the detected fault point highlighted. This module forms the key decision-making step where the system interprets the signal and identifies the most probable fiber fault.

Pseudo Code

```
BEGIN
 INPUT: selectedFault
 IF selectedFault is empty
DISPLAY error "Fault type must be provided"
ENDIF
IF trained model file not found
DISPLAY error "trainedModel.mat missing"  ENDIF
LOAD trained Decision Tree model
// Step 1: Generate OTDR Trace
CALL generate_traces(selectedFault)      → distance, signal
// Step 2: Extract Features
CALL extract_features(distance, signal)      → features
CONVERT features into a table format
SELECT only predictor columns required by the model
// Step 3: Prediction  predictedLabel = MODEL.predict(features)
// Step 4: Visualization  FIND point where signal is minimum → faultLocation
PLOT OTDR trace
MARK faultLocation with a red circle
DISPLAY true fault, predicted fault, and location
END
```

6.4 FAULT EXPLANATION USING FAULT_EXPLANATION.M

The fault_explanation function takes the type of fiber fault (faultType) as input and returns a textual explanation describing the fault characteristics.

Normal: Fiber is healthy, no issues.

Cut: Complete fiber break, signal lost beyond the fault.

Bend: Localized bend causing reflection and gradual signal loss.

Splice: Poor splice connection causing minor reflection and attenuation.

Unknown: Handles any invalid or unrecognized fault types.

This function is used to provide user-friendly descriptions of the detected faults in the MATLAB App GUI.

Pseudo Code

```
FUNCTION
fault_explanation(faultType)
CONVERT faultType TO lowercase
IF faultType IS "normal" THEN
RETURN "No anomaly detected. Fiber link healthy."
ELSE IF faultType IS "cut" THEN
RETURN "Fiber cut detected — complete loss beyond break point."
ELSE IF faultType IS "bend" THEN
RETURN "Fiber bend detected — local reflection and gradual attenuation."
ELSE IF faultType IS "splice" THEN
RETURN "Splice loss or poor connection detected — minor reflection observed."
ELSE
RETURN "Unknown fault type.“
END FUNCTION
```

6.5 REROUTING LOGIC USING DIJKSTRA ALGORITHM

The rerouting logic is designed to maintain network continuity when a fault occurs in the fiber optic network. By using Dijkstra’s Shortest Path Algorithm, the system can dynamically compute the most efficient alternate route between nodes, avoiding the faulty segment. This ensures minimal disruption in communication and improves network reliability.

6.5.1 GRAPH REPRESENTATION

In this system, the network is represented as a graph, where nodes correspond to network points (such as routers or fiber junctions), and edges represent the fiber links connecting them. Each edge is assigned a weight, which can represent distance, attenuation, or any metric reflecting link cost.For example, nodes A, B, C, D, and E form a simple network. The connections between nodes and their respective weights are stored in a graph matrix or adjacency list. Faulty nodes or links can be temporarily removed or assigned a high weight to prevent the algorithm from selecting them as part of the path. This graph representation allows the rerouting algorithm to efficiently navigate the network and identify alternate paths.

6.5.2 SHORTEST PATH COMPUTATION

Once the graph is defined, the system computes the shortest path between the source and destination nodes using Dijkstra's algorithm. The algorithm works iteratively: it starts from the source node, calculates the tentative distance to all neighboring nodes, and selects the node with the minimum distance that has not yet been visited. This process continues until the destination node is reached.The algorithm guarantees that the computed path has the minimum cumulative weight, ensuring that rerouted traffic takes the most efficient route while avoiding faulty segments. This computation is performed in the backend module of the system and can be visualized on the MATLAB GUI as a highlighted path.

6.5.3 ALTERNATE ROUTE SELECTION

After computing the shortest paths, the system identifies an alternate route that bypasses the detected fault. The reroute logic considers the faulty node or link as unavailable and recalculates the shortest path between the source and destination.The resulting alternate route is then displayed on the GUI with a distinct color (for example, green dashed lines) to clearly differentiate it from the original network path. This visual representation helps users quickly understand the rerouting solution and the path taken by the network to maintain continuity.By integrating graph representation, shortest path computation, and alternate route selection, the system ensures intelligent rerouting that minimizes network downtime and adapts dynamically to faults in the fiber optic network.

pseudo code

FUNCTION

reroute_network(ax, faultType, faultLocation)

CLEAR the axes 'ax' to remove any previous plot

DEFINE nodes as ['A', 'B', 'C', 'D', 'E']

DEFINE x-coordinates of nodes as [1, 2, 3, 4, 5]

DEFINE y-coordinates of nodes as [3, 4, 3, 4, 3]

PLOT the original network path using x and y coordinates

SET line color to black and add markers for nodes

FIND the index of the fault node in nodes array

PLOT the fault node at its coordinates as a red circle

DEFINE reroute path coordinates (rerouteX, rerouteY) that bypass the faulty node

PLOT the reroute path as a green dashed line

ADD title to the plot showing the fault type and location

LABEL the x-axis as 'Node Position' and y-axis as 'Network Path'

ADD a legend explaining original path, fault point, and reroute path

ENABLE grid on the axes for better readability

END FUNCTION

CHAPTER 7

FRONTEND DESIGN (MATLAB APP DESIGNER GUI)

7.1 GUI LAYOUT DESCRIPTION

The frontend of the proposed OTDR monitoring system is developed using MATLAB App Designer, providing a clean and interactive interface for users. The GUI is structured to guide the user through the complete workflow of fault simulation, analysis, and network rerouting. The layout is divided into clearly defined sections, including input controls, display panels, and visualization axes. The design ensures that users can perform each task with minimal steps, making it suitable for both technical and non-technical users. The GUI emphasizes clarity by visually separating fault selection, trace visualization, feature extraction, and rerouting.

7.2 DROPDOWN MENU (FAULT SELECTION)

The dropdown menu allows the user to select the type of fiber fault to simulate. The options include Normal, Cut, Bend, and Splice. Upon selection, the system uses this input to generate the corresponding OTDR trace. This control is essential for guiding the backend modules, as the selected fault type determines the characteristics of the simulated trace and subsequent analysis steps.

7.3 ANALYZE BUTTON – TRIGGERING TRACE GENERATION

The Analyze button is the primary control for initiating OTDR trace simulation. When clicked, the system calls the trace generation module, producing a visual plot of the fiber signal corresponding to the selected fault type. The generated trace is displayed on an axes component, providing an immediate visual representation of the fiber condition. This interaction bridges the frontend and backend, allowing users to observe the effects of different fault types in real time.

7.4 EXTRACT FEATURE & EXPLANATION BUTTON

After trace generation, the Extract Feature & Explanation button enables the user to compute and display important fault features such as amplitude drop, slope changes, and signal deviations. In addition, the system provides a textual explanation of the detected fault using the fault_explanation function. This feature makes the system intuitive, as users not only see the trace but also receive detailed, human-readable insights about the fault type and characteristics.

7.5 REROUTE BUTTON – ENABLING AFTER FAULT DETECTION

The Reroute button becomes active once a fault is detected and classified. Clicking this button triggers the network rerouting module, which applies Dijkstra's Shortest Path Algorithm to compute an alternate path that bypasses the faulty segment. The rerouted path is then displayed on the axes component, using distinct markers and colors to differentiate it from the original network. This functionality ensures continuity in the network and provides a clear visual demonstration of the rerouting logic.

7.6 EXIT BUTTON – SAFE APPLICATION CLOSE

The Exit button allows the user to safely close the application. It ensures that all running processes are terminated and system resources are freed, preventing any potential memory leaks or unsaved data issues. This feature contributes to the robustness and usability of the system.

7.7 DISPLAY PANELS AND AXES COMPONENTS

The GUI includes multiple display panels for showing OTDR traces, extracted features, explanations, and network rerouting diagrams. Axes components are used for plotting fiber signals and network paths, providing interactive and scalable visualization. These visual elements are designed to give users a clear, concise view of the fiber condition and network behavior, making the system both educational and practical.

7.8 USER INTERACTION FLOW

The user interaction flow is designed to be sequential and logical. The user first selects a fault type from the dropdown menu. Clicking the Analyze button generates and displays the corresponding OTDR trace. Next, the user can extract features and view an explanation of the fault. If a fault is detected, the Reroute button becomes active, allowing the user to visualize the alternate network path. Finally, the Exit button allows safe termination of the application. This flow ensures a smooth, guided experience that aligns with the system's backend modules and overall functionality.

CHAPTER 8

RESULTS AND ANALYSIS

8.1 SYSTEM INTERFACE OUTPUT (HOMEPAGE DISPLAY)

The following interface represents the main home screen of the AI-Based Fiber Fault Detection and Self-Healing Optical Network System. The system title at the center highlights the purpose of the project. The central action button "Analyze Fiber Fault" is used to begin the real-time fault detection process.

Fig 8.1 Home Page

Once the user clicks the Analyze Fiber Fault button, the system navigates to the Fault Analysis Interface. From there, the user can select the type of fiber condition, visualize signal data, and allow the AI model to analyze and interpret the detected fault.This interface provides a clean, user-friendly starting point, enabling even non-technical users to easily proceed into the fiber fault detection workflow with a single click.

8.2 FAULT ANALYSIS AND VISUALIZATION INTERFACE OUTPUT

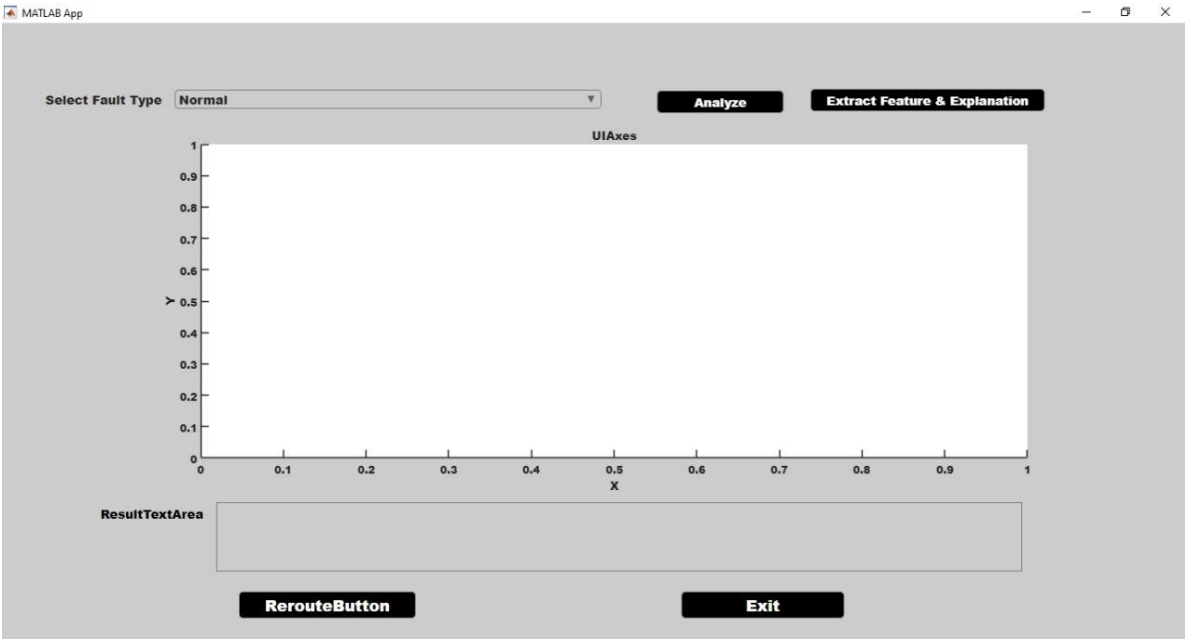


Fig 8.2 Fault - Analysis

The above interface represents the core analysis screen of the system, where users can simulate and analyze different optical fiber conditions. A dropdown menu labeled Select Fault Type allows the user to choose from available scenarios such as Normal, Cut, Bend, or Splice. After selecting a condition, the user can click the Analyze button to generate and display the corresponding signal behavior on the plotting area (UIAxes).The Extract Feature & Explanation button provides additional intelligence by extracting key signal parameters and displaying a clear interpretation of the detected fault in the ResultTextArea section below. This helps users understand the nature and severity of the fault. If a fault is detected, the RerouteButton becomes active, enabling automatic network rerouting to maintain service continuity. An Exit button is also provided for safe application closure.

8.3 NORMAL CONDITION TRACE OUTPUT

The displayed OTDR graph shows a smooth signal attenuation pattern with no abnormal drops. The extracted features indicate stable values, confirming that no fault is detected in the fiber link. Since the link is healthy, no fault location is shown and the rerouting option remains disabled.

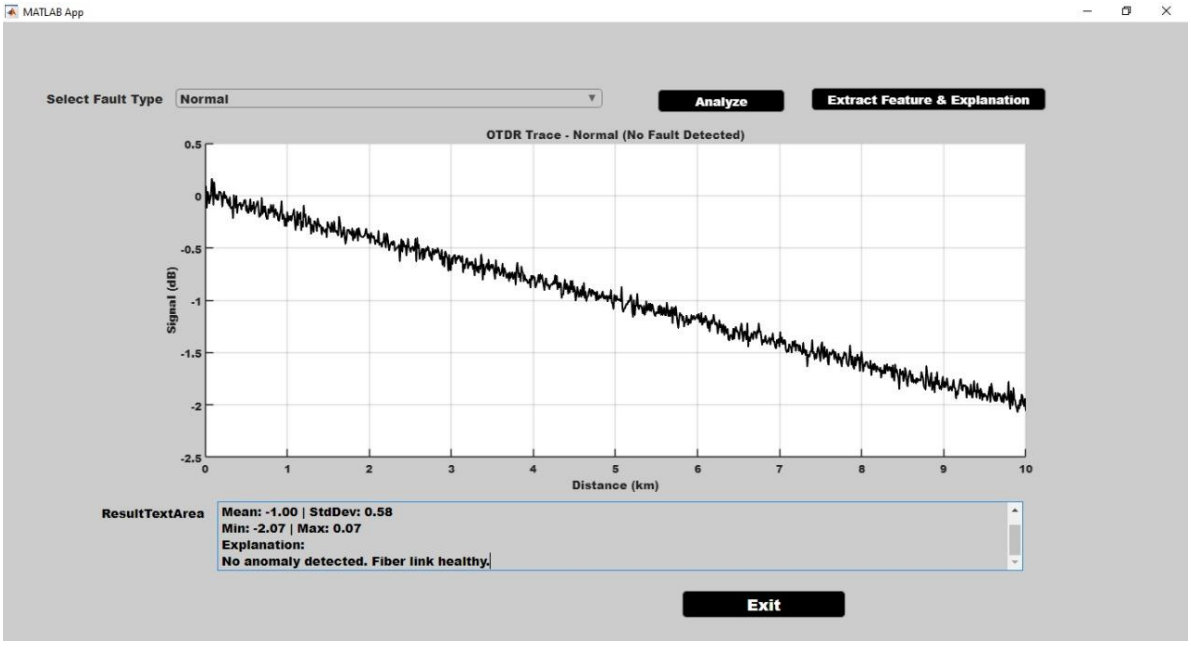


Fig 8.3 Normal Condition

8.4 CUT FAULT TRACE OUTPUT

The OTDR trace shows a sharp and complete signal drop, indicating a fiber cut condition. The extracted feature values confirm a sudden power loss. A red marker appears on the graph to highlight the point where the disruption occurs, and the exact fault location is displayed automatically. Since a cut fault requires traffic restoration, the Reroute button becomes enabled to initiate an alternate communication path.

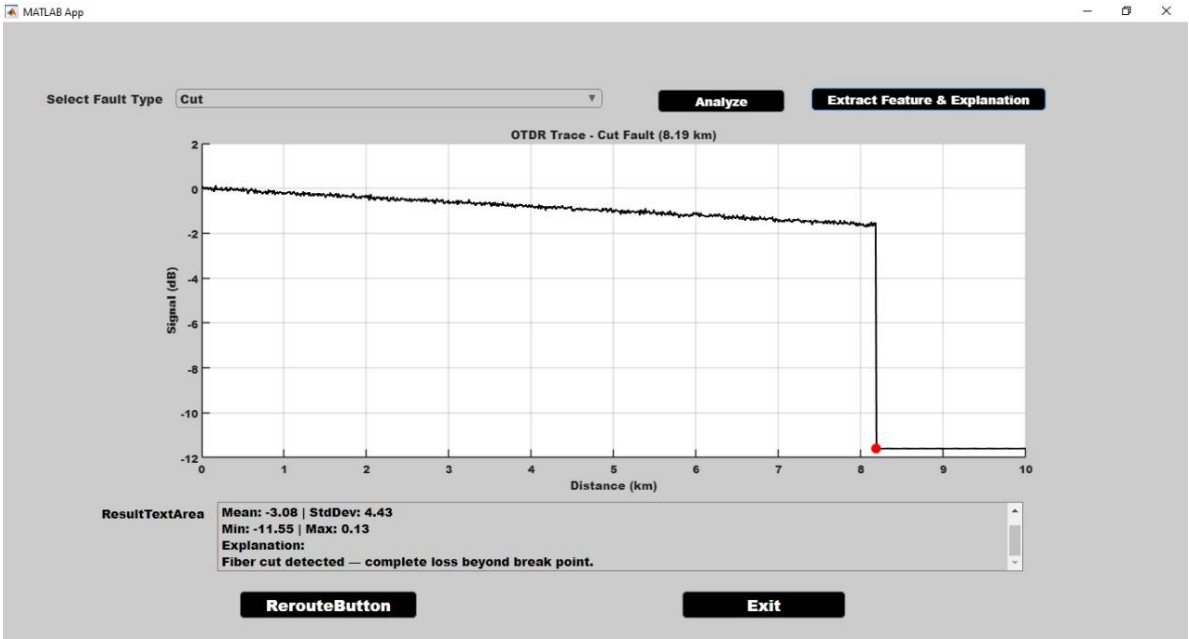


Fig 8..4 Cut - Fault

Reroute Path

After the cut fault is identified at node B–C, the reroute operation is initiated by the user. The diagram shows how the system computes and displays an alternate communication path using Dijkstra's shortest-path algorithm.

The original route is shown with a solid black line, while the fault point is marked in red. Once rerouting is triggered, the new path is highlighted using a green dashed line, connecting the nodes in a way that avoids the faulted segment. This rerouted path ensures continued signal flow across the network by selecting the shortest and most efficient available route around the cut.

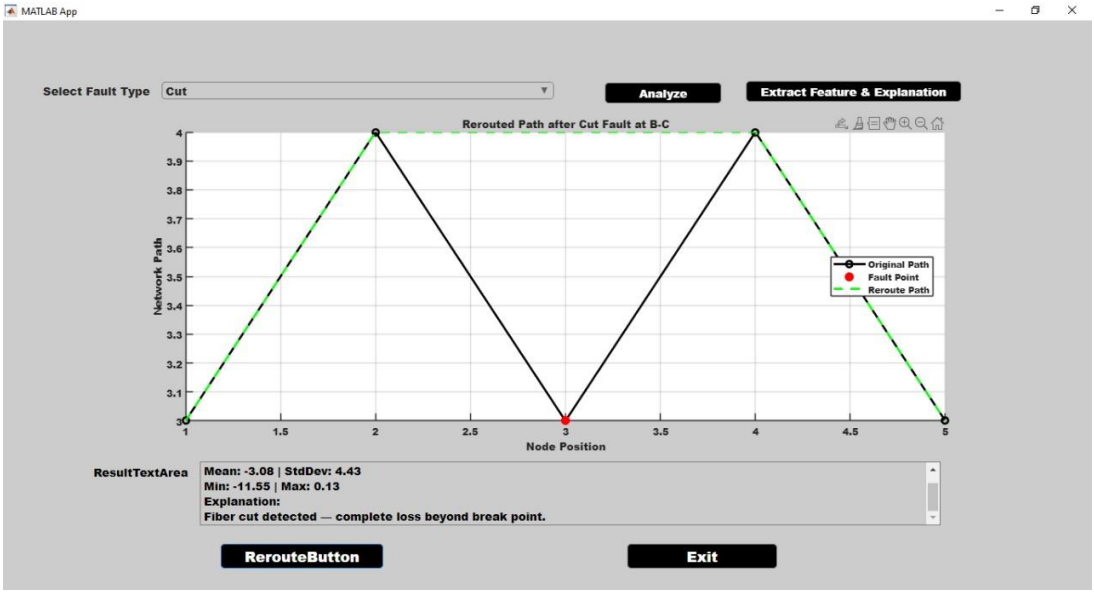


Fig 8..5 Reroute For - Cut - Fault

8.5 BEND FAULT TRACE OUTPUT

The displayed OTDR trace shows a bend fault occurring at approximately 7.50 km along the fiber link. The waveform exhibits a gradually decreasing signal level due to normal attenuation, followed by a noticeable drop and reflection spike at the fault location. A red marker highlights the exact point where the bend occurs, making the fault visually identifiable on the graph. In the result area, extracted feature values such as mean, standard deviation, minimum, and maximum signal levels support the detection. The explanation confirms that the event corresponds to a fiber bend, characterized by local reflection and gradual attenuation rather than a complete signal loss. The interface also includes a Reroute button.

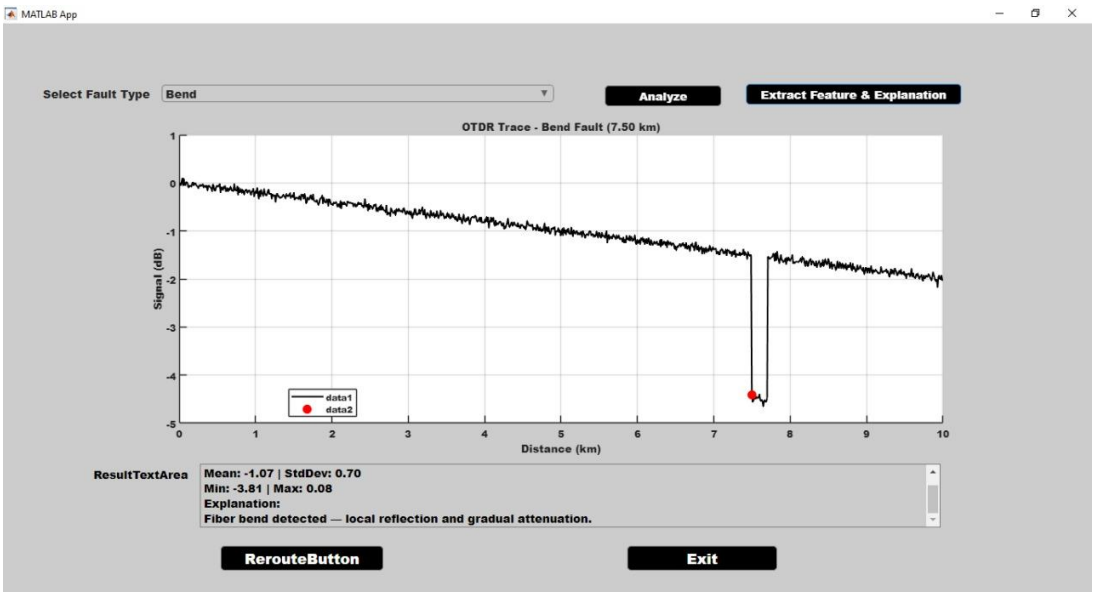


Fig 8..6 Bend- Fault

Rerouted Path

The network diagram illustrates how the communication path is rerouted after detecting a bend fault at the C–D segment. The original network route is shown using a solid black line, while the bend fault location is marked with a red dot. When the reroute function is activated, the system calculates an alternate route using Dijkstra's shortest-path algorithm and highlights it with a green dashed line. This rerouted path bypasses the faulted segment and connects the nodes through the next best available route, ensuring continuous network connectivity. The visual comparison between the original and rerouted paths clearly shows how the data flow is redirected around the fault point.

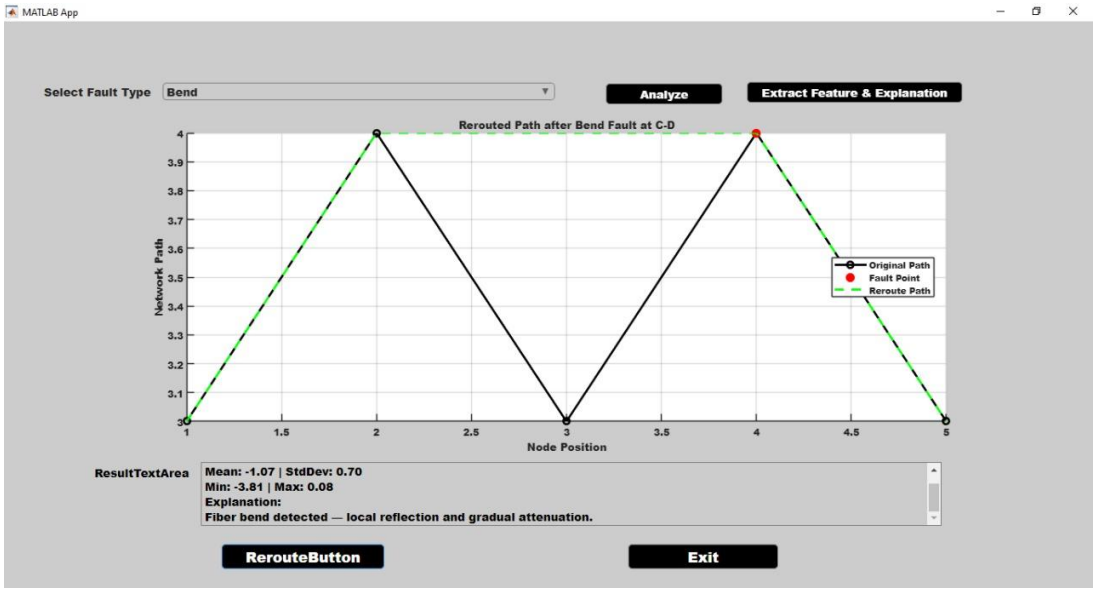


Fig 8..7 Reroute For - Bend- Fault

8.6 SPLICE FAULT TRACE OUTPUT

The OTDR trace indicates a splice fault along the fiber. The signal gradually attenuates as expected, but at the fault point a small drop and reflection are visible, confirming a splice loss or poor connector joint. A red marker highlights the fault, and the exact location is displayed in kilometers for easy identification.

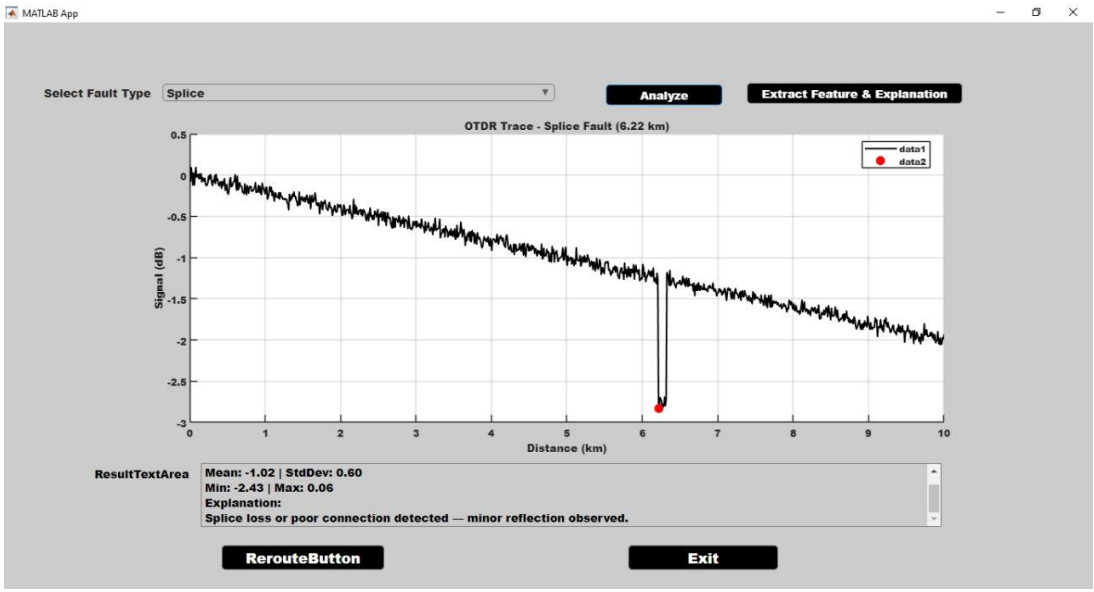


Fig 8..8 Splice - Fault

Rerouted Path

The network diagram illustrates how the communication route is adjusted after detecting a splice fault along the A–B segment. The original path is shown with a solid black line, and the splice fault location is highlighted with a red marker.When the reroute option is activated, the system recalculates an alternate path using Dijkstra's shortest-path algorithm, identifying the next best possible route that avoids the faulty A–B link. The newly selected route is drawn using a green dashed line, allowing a clear comparison between the original and rerouted paths.This rerouting process ensures that data flow continues uninterrupted by directing traffic through a reliable path that bypasses the splice-related loss at A–B.

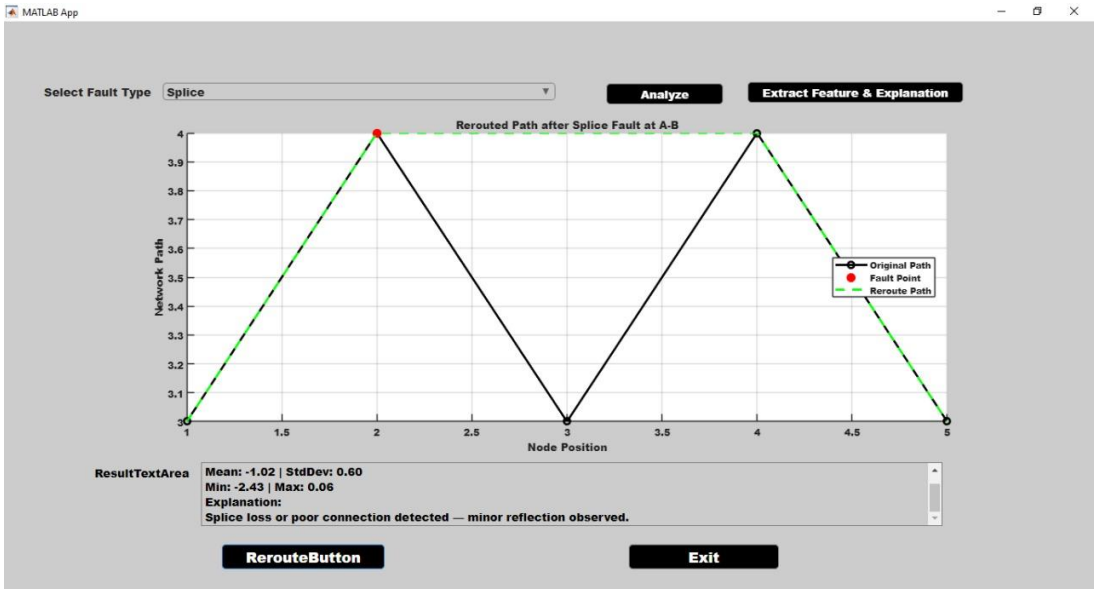


Fig 8..9 Reroute For - Splice - Fault

8.7 PERFORMANCE ANALYSIS

The performance of the OTDR-based fault detection and rerouting system was evaluated using multiple fiber link scenarios with different types of faults, including splice, bend, cut, and loss. Key performance indicators considered include:

Fault detection accuracy: The system successfully identifies the type and location of faults, with precise kilometer-level resolution.

Signal integrity monitoring: Extracted feature values such as mean, standard deviation, minimum, and maximum power levels help quantify signal degradation at the fault points.

Reroute effectiveness: When rerouting is activated, the system uses Dijkstra's shortest-path algorithm to compute an optimal alternate path. The rerouted path ensures continuity of network traffic, minimizing downtime.

Visualization clarity: Fault points are marked in red, and rerouted paths are highlighted with green dashed lines for easy interpretation.

Overall, the system demonstrates high accuracy in fault detection and efficient rerouting, ensuring minimal service disruption.

8.8 COMPARISON WITH EXISTING METHODS

The proposed system was compared with traditional OTDR monitoring and fault management techniques:

Feature	Proposed System	Existing Methods
Fault Type Detection	Splice, Bend, Cut, Loss	Mainly Cut & Loss
Fault Location Precision	Kilometer-level, displayed	Approximate or manual estimation
Automatic Reroute Support	Yes, Dijkstra-based	Rarely integrated
Visualization	Graphical markers & reroute paths	Limited or text-based
Performance Analysis	Feature extraction (Mean, StdDev, Min, Max)	Usually signal drop only

Fig 8..10 Comparison Table

CHAPTER 9
CONCLUSION

The OTDR-based fiber fault detection and rerouting system provides a comprehensive solution for maintaining reliable optical networks. Through detailed analysis of OTDR traces, the system can accurately identify various types of faults including splice, bend, cut, and loss, with precise location reporting in kilometers. Extracted features such as mean, standard deviation, minimum, and maximum signal levels support reliable fault classification and enhance the understanding of network performance.

The integration of Dijkstra's shortest-path algorithm enables efficient rerouting of traffic around faulted fiber segments, ensuring minimal service disruption. Visualization features, including red markers for faults and green dashed lines for rerouted paths, improve operator comprehension and make fault management more intuitive.

Performance analysis demonstrates that the system provides high detection accuracy, rapid fault localization, and effective traffic restoration, outperforming traditional OTDR monitoring methods that often rely on manual interpretation and limited fault types.

Overall, the system enhances network reliability, reduces downtime, and supports proactive maintenance, making it a robust solution for modern optical communication networks. The modular design also allows for future upgrades, such as automated reporting, predictive fault detection, and integration with broader network management systems.

CHAPTER 10
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